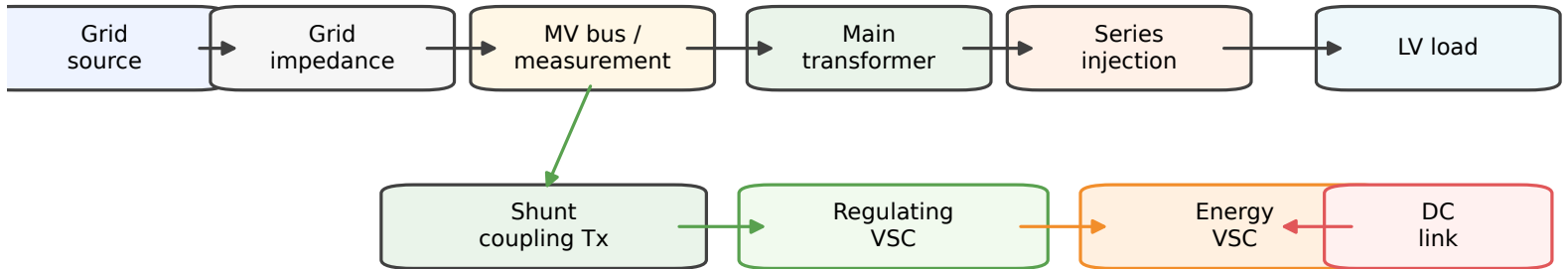


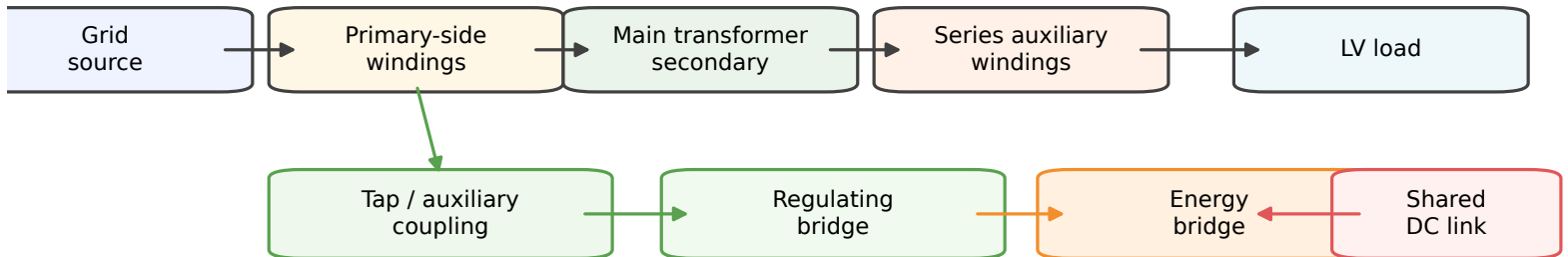
HPT switch-level topology and controller interface

Topology 1: shunt VSC plus series injection branch



Switch-level plant; controller commands are modulation references.

Topology 2: transformer-integrated regulating and energy branches



Schematic abstraction of the switch-level Simulink models used for validation.